Progress Report 6

1. **Project title:**
   Touchscreen interface for a multipurpose robot (R1D4) (for the first semester) and swarm of robots to complete locomotive and object-searching tasks (for the second semester).

2. **Names, student numbers, and emails**:
   Group 36 members: (all members are registered in ECSE 458-D1)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**

All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

Weekly meeting with the entire group: Wednesdays at 3:00 pm. Weekly meeting with the advisor: Thursdays at 4:30 pm

In addition to these scheduled meetings, this semester, we have decided to dedicate individual free time to working on the swarm of robots at Trottier 4130. This extra lab time will give us the opportunity to work directly with the hardware, identify and fix issues, and improve our system, helping us make steady progress outside of our regular meetings.

Recent progress

We completed the R1-D4 interface app. This includes adding persistent data storage, modifying the delivery data sending logic to send data only 5 minutes prior to the delivery, adding the update and delete functionalities. Finally we added the final UI frontend changes to make the app look cohesive and visually pleasing. The app was built and then installed on the Android tablet and can now be operated on the tablet simply by connecting the ESP32.

Additionally we also prepared for the mid-term presentation which included preparing the slides and rehearsing for the presentation itself.

We started getting acquainted with the SwarmBot project. We assessed the current condition of the various bots, and attempted to set up sending commands to the BoeBot. Unfortunately, we ran into multiple issues regarding the hardware (power system) and software (connecting to servers). We concluded that the HoverBot will be the focus of our remaining time, since this is the Swarm bot in the best condition.

Future plans

We hope to complete a rudimentary vision-based tracking system that can determine where the HoverBot is in the room, along with it's orientation and the distance from the goal location. This should be completed by March 17.

Then, we hope to set up the HoverBot to accept commands over a server. Finally, we plan to integrate HoverBot with the vision-based tracking system in order to send commands to the HoverBot based on the current location according to the tracking system, thus making a fully-functional and automatic navigation system for the HoverBot, with potential for scalability to multiple robots for future groups.

Lastly, we hope to write high quality documentation for our Swarm bot progress, along with visual guides in the form of posters to aid our presentation during design day.

Ethics and Equity

In our project, we did not encounter any ethical problems or equity issues. However, it is important to consider potential ethical problems that can be posed in engineering projects, especially in autonomous and swarm robotics.

For the R1-D4 hospital robot, some of the ethical considerations that will be taken into account include patient privacy, data security, and ensuring equal access to healthcare services. The robot will handle sensitive information, such as patient locations and medication details, which raises concerns about data protection. The design must also consider accessibility to ensure that it does not discriminate against some patients in favor of others due to mobility limitations or infrastructural limitations. We designed our touch screen with this information in mind, keeping our buttons large and our UI in general as intuitive as possible.

In the swarm robot project, there are not a lot of obvious ethical concerns since this project does not have a concrete real world application yet. Although we could imagine scenarios where these swarm robots were deployed in disaster relief settings. If they were coordinating deliveries to bring life saving goods, we would need to think about how to prioritize these deliveries and make sure the most ethical decision is being made. To be fully transparent, we recognize that this

example is a bit of a stretch but as our project does not have a direct ethical dilemma to deal with we had to think of a hypothetical scenario.

More broadly, robotic ethical issues generally involve the trade-off between automation and human oversight, addressing the biases of AI-driven decision-making, and bringing technological advancement to all users in a fair manner. Although we did not specifically face these issues in our project, being sensitive to these considerations allows us to develop responsible, fair, and effective engineering solutions.